

# Male Hormone Patterns & Phenotypes

Clinical pattern-recognition framework for testosterone optimization, metabolic interpretation, and male hormone assessment.

**Important Clinical Note:** These phenotypes are intended as pattern-recognition tools and educational frameworks — not standalone diagnoses. Interpretation should always incorporate symptoms, repeat morning testosterone testing when indicated, age, medications, sleep quality, metabolic health, fertility goals, and overall clinical context.

## Suggested Optimized Male Hormone & Metabolic Targets

| Marker                       | Suggested Functional / Optimized Range |
|------------------------------|----------------------------------------|
| Total Testosterone           | 700–1000 ng/dL                         |
| Calculated Free Testosterone | 120–220 pg/mL                          |
| Free Testosterone %          | 1.8–2.5%                               |
| Bioavailable Testosterone    | 300–500 ng/dL                          |
| SHBG                         | 20–40 nmol/L                           |
| Estradiol (Sensitive Assay)  | 20–35 pg/mL                            |
| LH                           | 3–8 mIU/mL                             |
| FSH                          | 2–8 mIU/mL                             |
| PSA                          | <1.5 ng/mL preferred in lower-risk men |
| Hematocrit                   | 42–50%                                 |
| Fasting Insulin              | 2–8 µIU/mL                             |
| hs-CRP                       | <1.0 mg/L                              |
| ApoB                         | <80 mg/dL                              |
| Triglycerides                | <100 mg/dL                             |
| HDL                          | >50 mg/dL                              |

### 1. Primary Testicular Failure Pattern

**Typical Labs:** Low total testosterone, low free testosterone, elevated LH/FSH

**Common Clinical Features:** Low libido, erectile dysfunction, fatigue, infertility, reduced muscle mass

**Interpretation:** Suggests impaired testicular testosterone production.

**Clinical Considerations:** Repeat AM testosterone, fertility goals, prolactin, iron studies, urology/endocrinology referral when appropriate.

## 2. Secondary / Central Suppression Pattern

**Typical Labs:** Low total testosterone, low free testosterone, low or inappropriately normal LH/FSH

**Common Clinical Features:** Fatigue, low drive, obesity, poor recovery, sleep issues

**Interpretation:** May reflect hypothalamic or pituitary suppression.

**Clinical Considerations:** Assess obesity, OSA, opioid use, stress, pituitary causes, prolactin, thyroid function.

## 3. Obesity / Low SHBG Functional Low-T Pattern

**Typical Labs:** Low total testosterone, normal or borderline free testosterone, low SHBG

**Common Clinical Features:** Visceral adiposity, insulin resistance, fatty liver pattern

**Interpretation:** Total testosterone may appear falsely low due to suppressed SHBG.

**Clinical Considerations:** Prioritize free testosterone interpretation and metabolic optimization.

## 4. High SHBG / Low Free Testosterone Pattern

**Typical Labs:** Normal total testosterone, low free testosterone, elevated SHBG

**Common Clinical Features:** Symptoms despite 'normal' total testosterone

**Interpretation:** Bioavailable androgen may still be insufficient.

**Clinical Considerations:** Review thyroid, liver status, nutrition, aging-related SHBG elevation.

## 5. Aromatization / Estrogen-Dominant Pattern

**Typical Labs:** Borderline or low testosterone with higher estradiol

**Common Clinical Features:** Fluid retention, gynecomastia, emotional lability, low libido

**Interpretation:** Increased aromatase activity often associated with adiposity.

**Clinical Considerations:** Address metabolic health and body composition before reflexively suppressing estrogen.

## 6. Over-Replacement / Supraphysiologic TRT Pattern

**Typical Labs:** High total/free testosterone, suppressed LH/FSH, elevated hematocrit

**Common Clinical Features:** Acne, irritability, insomnia, elevated BP, edema

**Interpretation:** May indicate excessive androgen replacement.

**Clinical Considerations:** Review dosing strategy, frequency, estradiol, CBC, fertility implications.

## 7. TRT Non-Responder / Symptom-Mismatch Pattern

**Typical Labs:** Normal or high testosterone but persistent symptoms

**Common Clinical Features:** Fatigue, low motivation, poor sleep, mood symptoms

**Interpretation:** Symptoms may not be androgen-driven.

**Clinical Considerations:** Evaluate sleep apnea, thyroid, mental health, metabolic dysfunction, nutrient deficiencies.

## 8. Fertility-Preservation Pattern

**Typical Labs:** Borderline testosterone with preserved or desired fertility

**Common Clinical Features:** Low libido or energy with concern about sperm production

**Interpretation:** Traditional TRT may suppress spermatogenesis.

**Clinical Considerations:** Discuss fertility goals prior to testosterone therapy.

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**ClinIQ Integration Idea:** These phenotypes work best when paired with AI-driven pattern recognition that evaluates symptoms, labs, metabolic markers, medications, sleep, and longitudinal trends simultaneously rather than relying on isolated testosterone values alone.